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Social Cause Problem Statements: Choose one.

Problem Statement 1:

City Infrastructure Dependency Mapping & Resilience Intelligence Platform

Overview

The **City Infrastructure Dependency Mapping Platform** is designed to help city planners and emergency responders understand how critical urban systems—such as power grids, water supply, transportation, telecommunications, and emergency services are interconnected. Since disruptions in one system can trigger cascading failures in others, the platform models and visualizes these cross-system dependencies. By analyzing how failures propagate across infrastructure networks, it enables stakeholders to anticipate risks, respond more effectively to disruptions, and make better decisions for improving urban resilience and infrastructure planning.

Objectives

1. Multi-Infrastructure Data Integration

Develop a system capable of integrating data from multiple infrastructure sectors, including:

- Power grid infrastructure
- Water supply and distribution systems
- Transportation networks
- Telecommunications infrastructure
- Emergency response services

The platform should organize this data into a **structured infrastructure dataset** representing system components and relationships.

2. Infrastructure Dependency Mapping

Model urban infrastructure as an interconnected network where:

- **Nodes** represent infrastructure components (e.g., substations, water pumps, hospitals).
- **Edges** represent dependencies between components or systems.

The system should visually represent these dependencies and allow users to explore interactions between infrastructure sectors.

3. Cascading Impact Analysis

Design mechanisms to analyze how disruptions in one component affect other systems.

The platform should help identify:

- Cascading failures across infrastructure networks
- Critical nodes whose failure can cause large disruptions
- Infrastructure sectors most vulnerable to cascading impacts

4. Scenario Simulation and What-If Analysis

Enable simulation of infrastructure disruption scenarios such as:

- Power grid outages
- Road network disruptions
- Telecom network failures
- Extreme weather impacts

The system should visualize how disruptions propagate across interconnected infrastructure systems.

5. Agentic AI-Assisted Infrastructure Insights

Implement an **Agentic AI system** that helps interpret infrastructure data and provide human-readable insights.
The agent should analyze infrastructure dependencies, identify potential cascading impacts, and assist planners or emergency responders in understanding system relationships.

The Agentic AI component can autonomously monitor infrastructure data, highlight critical risks, and summarize complex infrastructure interactions to support better decision-making.

6. Infrastructure Intelligence Dashboard

Develop an interactive dashboard that visualizes:

- Infrastructure network maps

- Dependencies between systems
- Cascading impact analysis
- Resilience indicators
- Simulated disruption scenarios

Expected Outcome

Teams should deliver a system capable of:

- Mapping dependencies across multiple urban infrastructure systems
- Visualizing interconnected infrastructure networks
- Analyzing cascading disruptions and identifying critical nodes
- Simulating infrastructure failure scenarios
- Providing AI-assisted explanations and insights
- Presenting infrastructure intelligence through an interactive dashboard

Problem Statement 2:

Circular Economy Marketplace for Reusable Materials

Overview

The **Circular Economy Marketplace Platform** is a digital system designed to connect individuals, businesses, and organizations that have surplus reusable materials; such as wood, metal, plastics, furniture, electronics, and construction materials; with others who can reuse them. Cities generate large amounts of waste, and many valuable materials end up in landfills due to the lack of an efficient system for redistribution. This platform aims to reduce waste, support sustainable resource use, and encourage circular economy practices by enabling communities to exchange, repair, and reuse materials instead of discarding them. It helps minimize environmental impact while providing affordable resources to small businesses and community organizations.

Objectives

Participants must design a platform that enables efficient exchange and reuse of surplus materials.

1. Reusable Material Listing and Inventory System

Develop a system that allows individuals, businesses, or organizations to list reusable materials they want to donate, sell, or redistribute.

Examples of materials may include:

- construction materials
- furniture and office equipment
- packaging materials
- electronics and appliances
- industrial surplus materials

The system should maintain a **structured inventory of reusable materials**.

2. Material Discovery and Marketplace Interface

Users should be able to explore available materials through an intuitive marketplace interface.

The platform should support:

- search and filtering by material type
- location-based discovery
- material availability tracking
- browsing by categories such as construction, electronics, packaging, and furniture

3. Intelligent Matching Between Supply and Demand

Participants should develop mechanisms that help match material suppliers with potential users.

Possible features may include:

- recommending relevant materials to users based on their needs
- suggesting nearby organizations that could reuse listed materials
- identifying industries or communities where certain materials are commonly reused

Basic machine learning or rule-based systems can be used to improve **resource matching efficiency**.

4. Material Lifecycle and Sustainability Insights

The platform should provide insights on how material reuse contributes to sustainability.

Possible features may include:

- estimating waste diverted from landfills
- calculating approximate environmental benefits of material reuse
- tracking how many times materials are reused or redistributed

These insights can encourage users to participate in circular economy initiatives.

5. Logistics and Redistribution Coordination

One major challenge in material reuse is **transportation and logistics coordination**.

Participants should design mechanisms that support:

- coordination between suppliers and receivers
- pickup and delivery scheduling
- collaboration with volunteer groups or logistics partners

The system should help reduce the friction involved in redistributing materials.

6. AI Sustainability Assistant (LLM / SLM)

Participants are encouraged to implement a lightweight **LLM or Small Language Model (SLM) powered assistant** that helps users understand reuse opportunities and sustainability benefits.

The assistant can interact with users in natural language while also using a **light agentic workflow** to analyze marketplace data, identify relevant material listings, and recommend reuse opportunities.

Example queries may include:

"Where can I find reusable construction materials nearby?"
"What can be done with surplus wooden pallets?"
"Which organizations might need these materials?"

The system should convert marketplace data into practical reuse suggestions and sustainability insights.

7. Circular Economy Impact Dashboard

Develop a dashboard that visualizes platform activity and sustainability impact, such as:

- volume of materials redistributed
- types of materials most frequently reused
- environmental impact metrics
- participation across communities or industries

Expected Outcome

Teams should develop a system capable of:

- listing and managing reusable materials
- enabling a marketplace for material exchange

- matching supply and demand for reusable resources
- coordinating redistribution logistics
- providing sustainability insights and impact metrics
- assisting users through an AI-powered guidance system
- visualizing circular economy activity through an interactive dashboard

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Problem Statement 3:

Governance Intelligence & Political Accountability Platform

Overview

Political parties present manifestos outlining promises to citizens during elections, but it is often difficult to track how these commitments translate into legislation, policies, or government actions. Legislative processes such as bills, amendments, and policy announcements are complex and scattered across multiple sources, making them difficult for citizens to interpret.Your challenge is to build a Governance Intelligence & Political Accountability Platform that aggregates information from sources such as political manifestos, legislative documents, policy announcements, and public news data to analyze how political commitments evolve into governance outcomes.Participants may integrate intelligent analysis tools using LLMs, SLMs, MLMs, or other machine learning techniques to interpret policy documents, identify relationships between policies, and assist users in understanding governance developments.

Objectives

1. Governance Data Aggregation

Develop mechanisms to collect governance-related information from multiple sources.

Possible Data Sources

- Election manifestos and policy documents
- Parliamentary or legislative records
- Government announcements or policy updates
- Public news sources and verified information portals

- Open government data platforms or relevant APIs

Possible Implementation Approaches

Participants may use techniques such as:

- Web scraping
- Data pipelines
- External APIs

These techniques should help gather, clean, and structure governance data from diverse sources.

2. Manifesto & Policy Commitment Repository

Develop a system that organizes political manifesto promises.

Possible Capabilities

- Categorizing promises by policy sectors
- Structuring promises into measurable commitments
- Linking promises to governance domains

3. Legislative Document Intelligence

Enable users to explore legislative materials such as:

- Bills and amendments
- Parliamentary discussions or summaries
- Policy announcements and regulations

Platform Objective

The platform should structure these documents so that they become easier to:

- Analyze

- Compare
- Search and interpret

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4. Promise–Policy Mapping Engine

Design mechanisms that connect election promises with legislative actions.

Possible Capabilities

- Identifying which policies or bills correspond to manifesto commitments
- Linking policy actions to election promises
- Highlighting promises with little or no policy progress

5. Legislative Evolution Tracking

Track how policies evolve during the legislative process.

Possible Capabilities

- Monitoring amendments to bills
- Identifying changes in policy scope
- Tracking updates to legislative proposals

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6. Governance Accountability Analysis

Design mechanisms to evaluate the progress of political commitments.

Possible Capabilities

- Identifying promises that have:
- Analyzing how frequently legislative actions correspond to manifesto commitments
- Estimating implementation levels of policy commitments

7. Policy Relationship & Similarity Analysis

Provide mechanisms to analyze relationships between policy documents.

Possible Capabilities

- Identifying similar or overlapping legislative proposals
- Detecting repeated or modified policy initiatives
- Grouping policies by thematic similarity

8. Policy Timeline & Governance Visualization

Design visual tools to explore governance developments.

Possible Visualization Methods

- Timelines showing policy evolution and legislative milestones
- Networks linking promises, policies, and legislative actions
- Clusters of related policy initiatives

9. Intelligent Policy Interpretation

Participants may integrate intelligent tools to assist users in understanding governance data.

Possible Capabilities

- Simplifying complex policy documents
- Summarizing legislative debates or bill descriptions
- Highlighting important policy developments
- Answering user queries related to governance activities

Possible Technologies

These capabilities may be implemented using:

- Large Language Models (LLMs)
- Small Language Models (SLMs)
- Masked Language Models (MLMs)
- Retrieval systems
- Other NLP and machine learning techniques

Expected Outcome

Teams should deliver a **prototype platform** capable of:

- Aggregating governance information from multiple sources
- Organizing manifesto commitments and legislative documents
- Analyzing relationships between election promises and policy actions
- Tracking legislative evolution and governance progress
  Visualizing policy timelines and governance relationships
- Assisting citizens in understanding democratic processes through intelligent analysis tools

Final Objective

The final solution should demonstrate how **technology can improve:**

- Transparency
- Accountability
- Public understanding of governance systems

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Problem Statement 4:

Global Conflict Impact Intelligence Platform for Public Awareness and Decision Support

Overview

The **Global Conflict Impact Intelligence Platform** is designed to analyze how major geopolitical events—such as wars, trade disputes, and diplomatic tensions—affect critical global systems like energy supply chains, transportation routes, economic stability, and public sentiment. During such events, information spreads quickly across news media, social platforms, and informal networks, often mixing verified facts with speculation and misinformation. This platform aims to process multiple information signals to assess real-world impacts, identify misleading narratives or exaggerated fears, and provide structured insights. By focusing on systemic analysis rather than simple news aggregation, it helps citizens, businesses, and policymakers better understand the true consequences of global conflicts.

Objectives

Participants must design a system capable of analyzing **geopolitical events and their broader impacts across interconnected domains**.

1. Multi-Source Information Integration

Develop a system that integrates data signals from multiple domains such as:

- global news and verified reporting sources
- economic indicators and commodity prices
- shipping and trade route activity
- energy supply and production data
- transportation and logistics networks
- public discourse signals from open platforms

2. Event and Narrative Extraction

Design mechanisms to detect **major geopolitical events** and extract **emerging narratives** surrounding them.

The system should be capable of identifying:

- key global events or conflicts
- dominant public narratives and concerns
- conflicting or competing interpretations of events
- rapid shifts in public discourse

3. Systemic Impact Analysis

Develop models that analyze how geopolitical events may influence interconnected systems such as:

- energy supply chains
- global trade routes
- commodity markets
- transportation and logistics networks
- regional economic stability

4. Risk and Impact Forecasting

Participants should design predictive models that estimate **potential impacts of geopolitical events** on critical systems.

Possible predictions may include:

- disruptions to global shipping routes
- changes in energy supply availability
- commodity price volatility
- trade route congestion risks

5. Narrative vs Reality Assessment

One of the key challenges during global crises is **distinguishing between perceived risks and actual systemic risks**.

Participants should develop mechanisms to:

- compare circulating narratives with measurable system indicators
- identify exaggerated or misleading interpretations of events
- highlight verified signals that confirm or contradict common assumptions

6. Agentic AI Geopolitical Intelligence Assistant

Participants should develop an **Agentic AI component** that helps interpret system outputs and provide clear explanations.

The agent should be capable of:

- monitoring system analytics across domains
- identifying emerging risk patterns
- summarizing key developments related to geopolitical events
- explaining complex global system interactions in simple terms

Example queries may include:

- “How could this conflict affect global oil supply?”
- “Are shipping routes currently disrupted due to this event?”
- “What economic sectors may be most affected by this conflict?”
- “Is the perceived risk supported by available data?”

7. Global Systems Intelligence Dashboard

Develop an **interactive dashboard** that visualizes:

- major geopolitical events and regions involved
- global trade and shipping routes
- energy supply and commodity indicators
- narrative trends and public concerns
- predicted systemic impacts

Possible Data Sources

Participants may use publicly available datasets such as:

- global shipping and maritime activity datasets
- commodity and energy market datasets
- economic indicator datasets
- open-source geopolitical event datasets
- public discourse datasets

Synthetic scenarios may also be generated to **test system behavior during simulated geopolitical crises**.

– Expected Outcome

Teams should develop a **prototype system** capable of:

- analyzing geopolitical events and global information signals

- identifying emerging narratives and public concerns
- mapping relationships between events and global systems
- forecasting potential impacts on supply chains, energy, and trade
- distinguishing between perceived risks and measurable impacts
- providing insights through an **Agentic AI intelligence assistant**
- visualizing global system dynamics through an **interactive dashboard**